RISE 2026 · SC26 · LIGHTNING TALK

Algorithms as if Architecture Mattered: Moving Less, Computing Faster

Rabab Alomairy

Ibn Rushd Assistant Professor, KAUST

Tuwaiq Intelligent Computing Lab (TICL)



IBN RUSHD

KAUST

تحكم
TAHAKOM

BACKUP · HISTORY

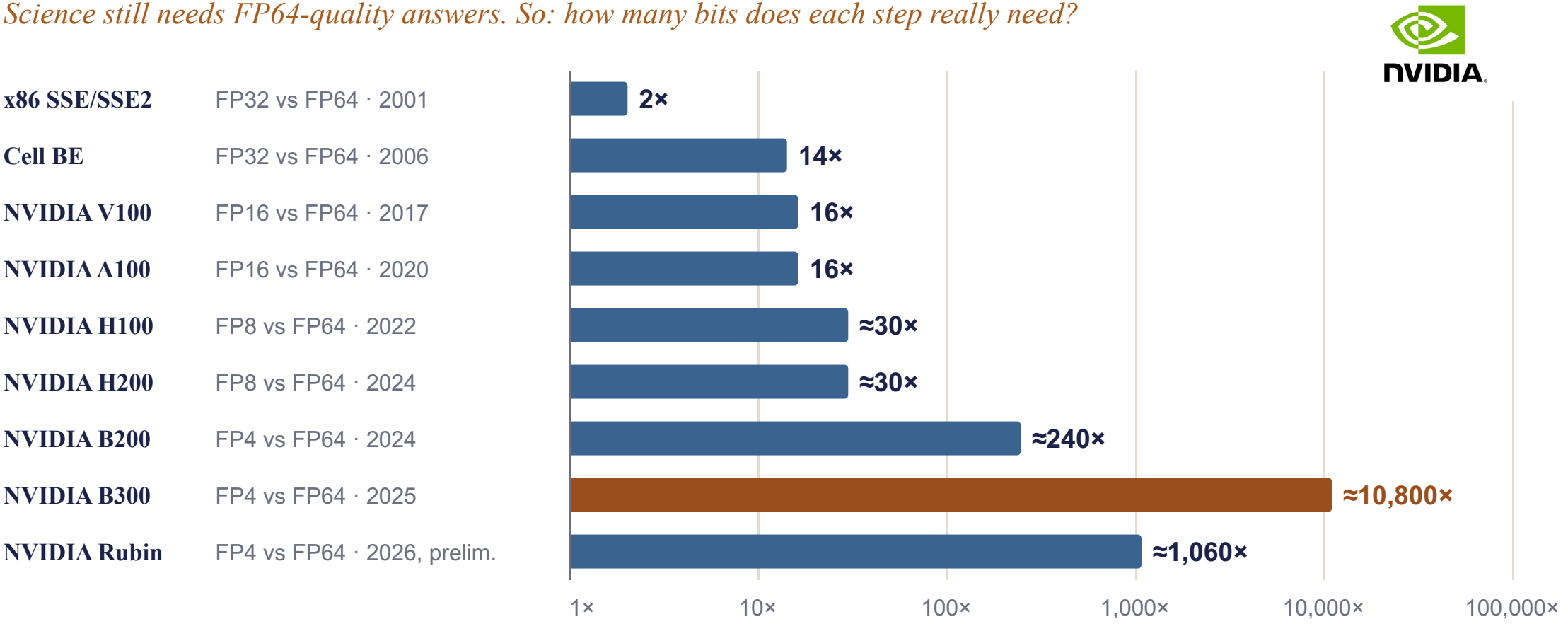
Seventy-five years of refinement: each motivation was set by where the hardware is going.

MOTIVATION	ACCURACY	STABILITY	SPEED			RECOVERY
ALGORITHM	Wilkinson: residuals in double length, solves in single	Fixed-precision IR stabilizes solvers (Skeel; Higham)	Langou et al.: fp32 LU, fp64 refinement, ≈8× on the Cell	GMRES-IR; IR in three precisions; tensor-core LU-IR	HPL-AI (HPL-MxP); GMRES-IR5; tiled mixed precision	FP64 emulation: Ozaki I in cuBLAS, Ozaki II; nested recursive precision
HARDWARE	**1948** ACE design (32-bit words, 64-bit product) Short words, but a free double-length accumulator.	**1977–91** Extra precision no longer cheap or portable	**2006** x86 SSE/SSE2: fp32 2× fp64. Cell BE: fp32 14× fp64	**2017–18** P100 native fp16; V100 tensor cores: fp16 16× fp64	**2019–21** V100 Summit: 3× HPL; A100: TF32, BF16. Fugaku A64FX: FP16 on CPUs Tensorcore FP64	**2024–26** B300: FP64 cut to ≈1.2 TFLOPS; Rubin: native FP64 33 TFLOPS FP64 Emulation 200 TFLOPS AMD MI430X: ~288

THE PROBLEM

Hardware now pays out in low precision, not FP64

Science still needs FP64-quality answers. So: how many bits does each step really need?



Bars: peak of each chip's lowest floating-point format ÷ peak FP64, per GPU (dense vendor peaks; FP64 = faster of vector and tensor; Rubin preliminary). Largest gap highlighted.

Each block gets only the bits it needs

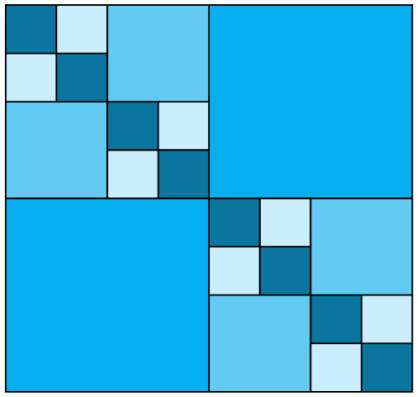
(Carrica et al., Euro-Par 2026)

Start from the precisions the hardware offers

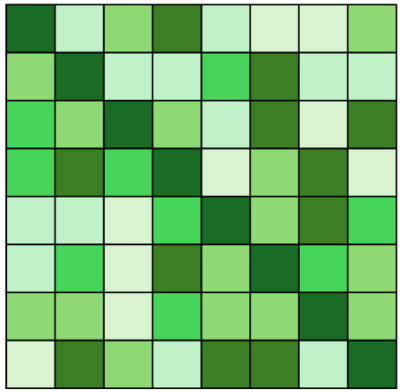
$$U = u_1, u_2, \ldots, u_N$$

Three steps

1. Score each block by its norm
2. Give each tile or tree level the lowest safe precision
3. Keep the global error below a chosen bound



Recursive (HODLR) mixed precision



Tile-based mixed precision

$$\xi_k = \max \frac{\|H_{ij}^{(k)}\|_F}{\|H\|_F}$$

$$u_k \leq 2^{-k/2} \frac{\epsilon}{\xi_k}$$

$$\|\hat{H} - H\|_F \lesssim \epsilon \, (2\sqrt{2}\ell + 1) \, \|H\|_F$$

Erin Carson (2025)

$$\|A_{ij}\|_F < u_{high} * \|A\|_F / (NT * u_{low})$$

$$\|\hat{A} - A\|_F \leq u_{high} \|A\|_F$$

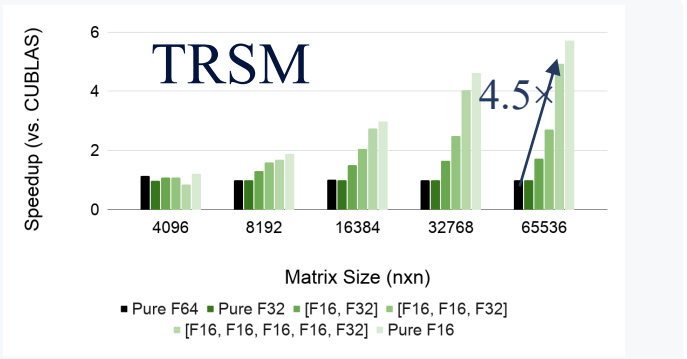
Nick Higham (2022)

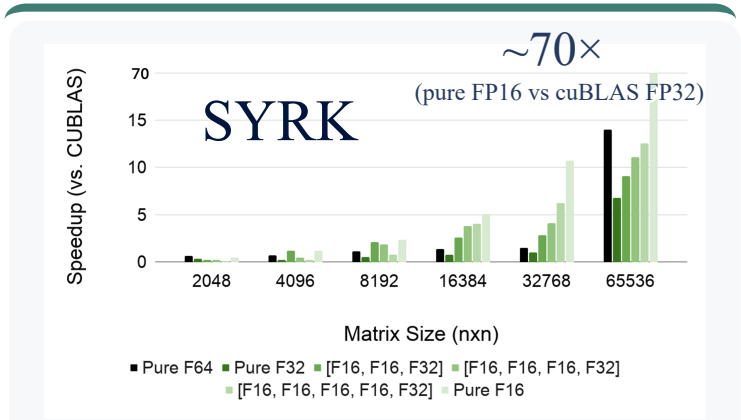
Award-level science on tile-based mixed precision

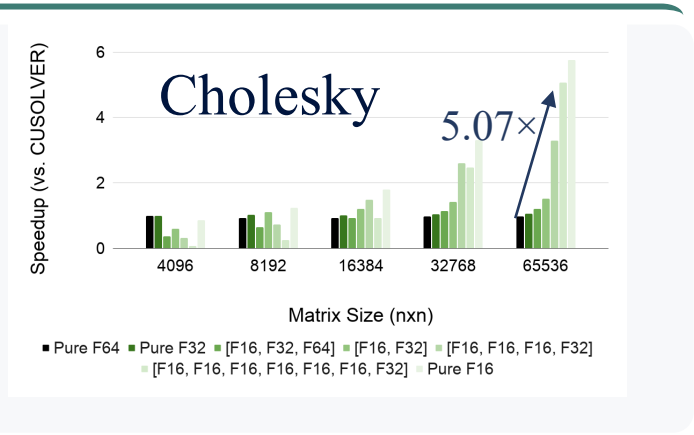
2024 ACM Gordon Bell Prize for Climate Modelling (KAUST climate emulator) · 2024 ACM Gordon Bell Prize finalist (our mixed-precision KRR for GWAS)

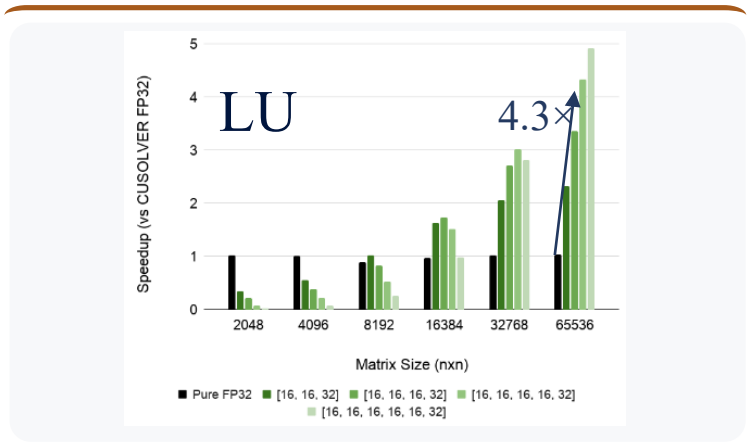
RESULTS · ONE NVIDIA H200 GPU

~5× faster computation, at an accuracy you choose









Setup

1 × NVIDIA H200 (also AMD MI300X)

Recursive kernels in Julia

Baselines: cuSOLVER, cuBLAS

n up to 65,536

Accuracy $\|A - LL^T\| / \|A\|$

Pure FP64: > 15 digits

[F16, F32, F64]: ≈ 9 digits

[F16, F16, F16, F32]: ≈ 5–6 digits

Pure FP16: < 4 digits

Carrica et al., Euro-Par 2026

RESULTS · AT SCALE · SUSTAINABILITY

Up to 72% less energy on Shaheen III

Cholesky of a 3.24M × 3.24M Matérn 2D covariance matrix · 512 CPU nodes (AMD EPYC Genoa) · PaRSEC runtime

Dense FP64

2,020 MJ

4,746 s

full double precision

Dense + MP

1,065 MJ

2,650 s

−47% energy · −44% time

TLR + MP

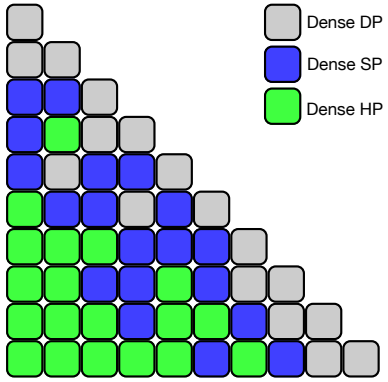
575 MJ

1,471 s

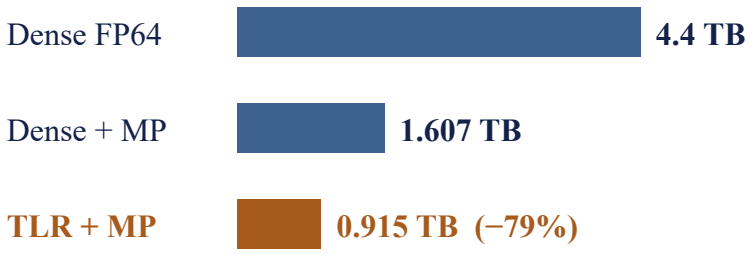
−72% energy · −69% time



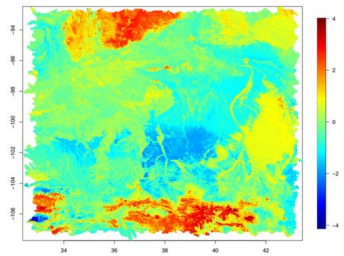
Shaheen III, KAUST



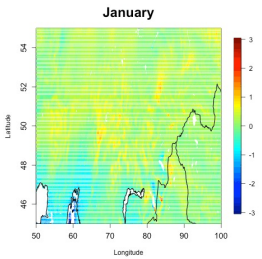
Memory footprint (~1M-point problem)



Target applications



Soil moisture, Mississippi basin



Evapotranspiration, Central Asia

SAME IDEA FOR AI · LLM INFERENCE

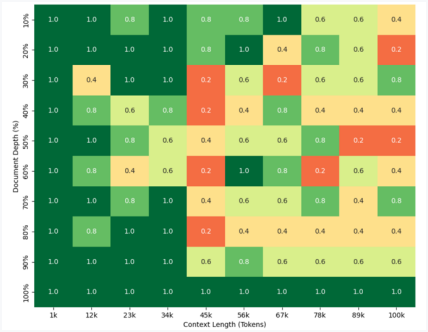
Same idea for AI: outlier-aware 2-bit KV cache

A few channels carry the outliers: give them 4 bits, the rest 2 bits · Llama-3.1-8B, needle-in-a-haystack to 100K tokens

Uniform 2-bit

71.4%

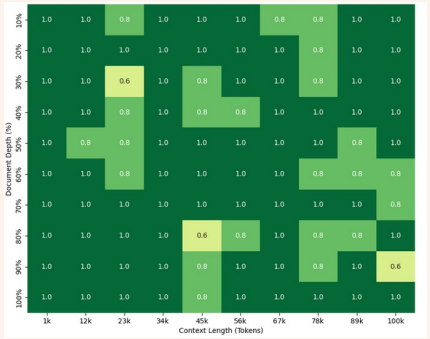
2.0 bits · 1.07 GB



Ours: outliers at 4-bit

94.2%

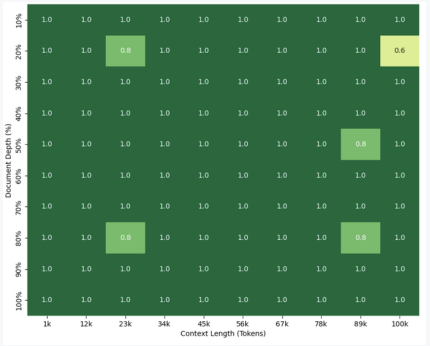
≈2.2 bits · 1.18 GB



Uniform 4-bit

98.5%

4.0 bits · 2.14 GB



≈2.2

average bits per KV-cache entry

–45%

GPU memory vs uniform 4-bit

~4%

extra quantization time

Each cell: needles found (of 5) by context length (x) and needle depth (y). Dark green = all found.

Team: Waad Alahmed, Shahad Alfawzan, Deema Alresheed, Shaden Alsalhy, Lina Alyahya

Accepted: SLM-Agents workshop, NeurIPS 2026

TAKEAWAY

Use only the bits each step needs

≈5×

faster computation on one GPU

precision set per recursion level (NVIDIA H200)

−72%

energy at scale

3.24M Cholesky on 512 Shaheen III nodes

≈2.2 bits

per entry in LLM KV caches

45% less memory than 4-bit, near-4-bit accuracy

Collaborate

Bring your application

if it needs FP64-quality answers at scale

Join

Hiring: PhD, MS, postdocs

research on Shaheen III with MIT, ORNL and ICL partners

Connect

Find me after the session

happy to share papers, code and ideas

In collaboration with